JAVA CODING FOR KIDS

Learn by doing 10 ready to type game in java

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1 Setting up for fun.

Learning to code is most fun when you're creating something you can play with! By typing and building these simple games, you're not just learning Java programming concepts—you're also developing problem-solving skills and understanding how code works in real-world applications.

The best way to learn is by doing. As you type each line of code, you'll see how it all comes together to create a working game. You'll learn about variables, loops, and conditional statements in a hands-on way that makes sense. Plus, you can modify the game to make it your own, which is a great way to experiment and learn even more!

So, get started and build your own games. You'll be amazed at how quickly you can create something cool and fun!

How to get started?

If you have already an IDE you can use that or you can type your games online and play without installing extra software.

Here is two simple path you can take to get ready:

1. Online Fun

Open a Web Browser:
Use a browser like Google Chrome, Firefox, or Safari.

Search for "Online Java Compiler":
Type these exact words in the search bar and press Enter.

Choose a Compiler:
Look at the search results and pick a compiler that looks easy to use. Some popular options include:

JDoodle: Known for its simplicity and ease of use.

TutorialsPoint: Offers a comprehensive environment with features like code execution and sharing.

OneCompiler: Provides a robust editor with features like auto-completion and error checking.

2. Use an IDE

If you have the possibility to set up an IDE (Integrated Development Environment) locally with the help of a parent or friend, you should try that! IDEs like IntelliJ IDEA or Eclipse make coding easier and provide powerful tools to help you learn Java.

However, if setting up an IDE feels too complicated right now, don’t worry! You can start coding online using one of the compilers mentioned above. It’s simple, quick, and perfect for beginners.

```
import java.util.Scanner;
import java.util.Random;

public class CatchTheStar {
  public static void main(String[] args) {
    Scanner scanner = new Scanner(System.in);
    Random random = new Random();

    // Game settings - kids can modify these values
    int gameWidth = 10;      // Width of the game area
    int playerPosition = 5;   // Starting position (middle)
    int score = 0;           // Starting score
    int rounds = 15;         // Number of rounds to play
    char playerChar = '^';    // Character representing the player
    char starChar = '*';      // Character representing stars
    char obstacleChar = 'X';  // Character representing obstacles

    System.out.println("=== CATCH THE STAR ===");
    System.out.println("Move left (L) or right (R) to catch stars (*) and avoid obstacles (X)");
    System.out.println("Stars = 10 points, Obstacles = -5 points");
    System.out.println("Press Enter to start the game!");
    scanner.nextLine();

    // Main game loop
    for (int round = 1; round <= rounds; round++) {
      // Create a falling object (either star or obstacle)
      int objectPosition = random.nextInt(gameWidth) + 1;
      boolean isStar = random.nextBoolean();
      char fallingObject = isStar ? starChar : obstacleChar;

      // Display the falling object
```

```
System.out.println("\nRound " + round + "/" + rounds + " | Score: " + score);
System.out.println();

// Show the falling object row
for (int i = 1; i <= gameWidth; i++) {
   System.out.print(i == objectPosition ? fallingObject : " ");
}
System.out.println();

// Show the player row
for (int i = 1; i <= gameWidth; i++) {
   System.out.print(i == playerPosition ? playerChar : " ");
}
System.out.println();

// Get player move
System.out.print("Move (L/R): ");
String move = scanner.nextLine().toUpperCase();

// Process player movement
if (move.equals("L") && playerPosition > 1) {
   playerPosition--;
} else if (move.equals("R") && playerPosition < gameWidth) {
   playerPosition++;
}

// Check if player caught the object
if (playerPosition == objectPosition) {
   if (isStar) {
      System.out.println("You caught a star! +10 points");
      score += 10;
   } else {
      System.out.println("Oh no! You hit an obstacle! -5 points");
```

```
            score -= 5;
          }
        } else if (isStar) {
          System.out.println("You missed a star!");
        } else {
          System.out.println("You avoided an obstacle!");
        }
      }

      // Game over - display final score
      System.out.println("\n=== GAME OVER ===");
      System.out.println("Final score: " + score);

      // Rate the performance
      if (score >= 100) {
        System.out.println("Amazing! You're a star catcher champion!");
      } else if (score >= 50) {
        System.out.println("Great job! You're getting good at this!");
      } else if (score >= 0) {
        System.out.println("Not bad! Keep practicing!");
      } else {
        System.out.println("Better luck next time!");
      }

      scanner.close();
    }
  }
```

What kids can easily modify:

- Change the game width to make it easier or harder
- Change the characters used for the player, stars, and obstacles
- Adjust the scoring system (how many points for stars/obstacles)
- Change the number of rounds
- Modify the final messages based on score
- Add new features (like different types of falling objects)

The game uses only basic Java concepts (variables, loops, conditionals, input/output), making it accessible for beginners. Kids can run this in any Java environment, including online IDEs if they don't have Java installed locally.